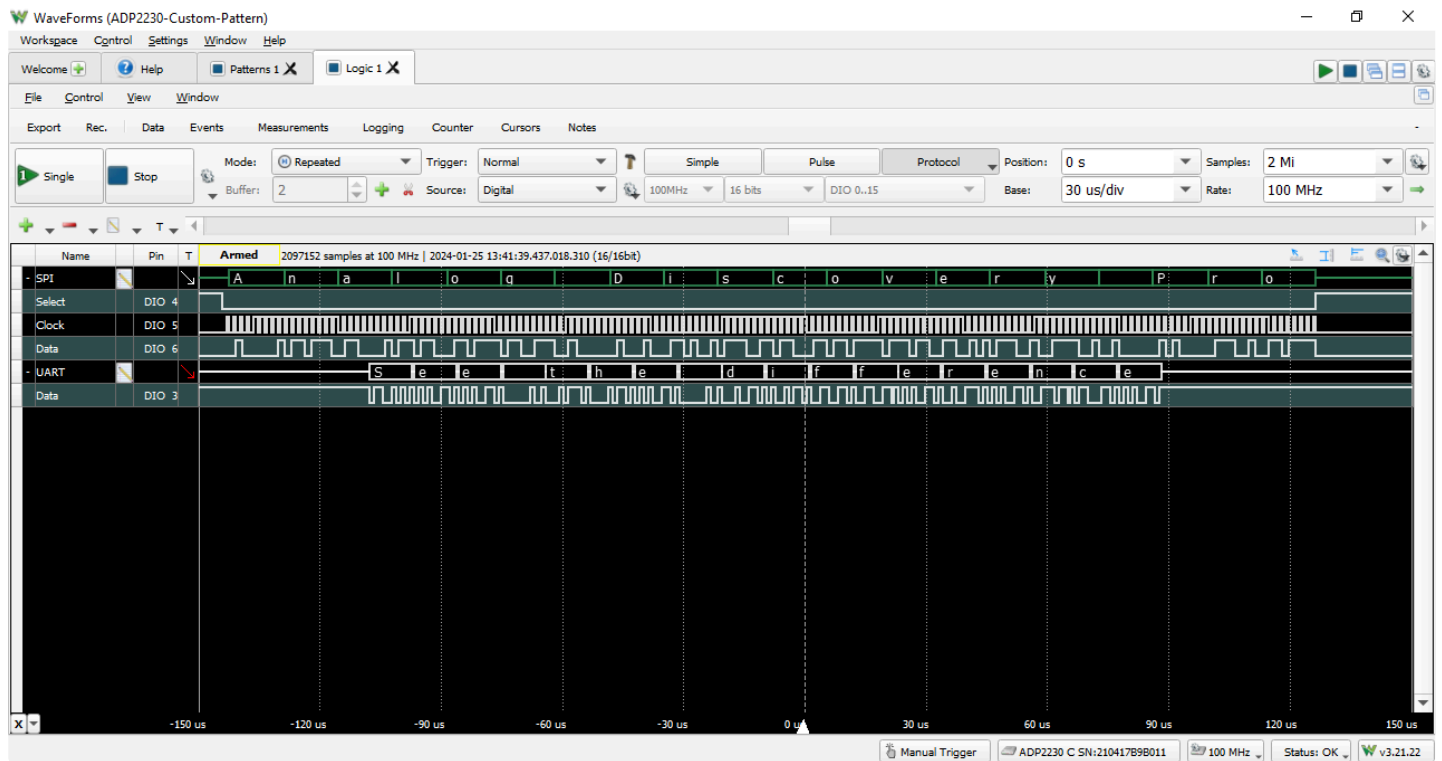


Using the Logic Analyzer



Introduction

Logic Analyzers are used to capture and view digital signals. This guide explains the use of WaveForms' *Logic Analyzer* instrument. The instrument is also referred to as *Logic* in WaveForms.

Prerequisites

- A Diligent Test & Measurement Device with Digital Inputs:
 - Analog Discovery 3 (<https://diligent.com/reference/test-and-measurement/analog-discovery-3/start>)
 - Analog Discovery Studio Max (<https://diligent.com/reference/test-and-measurement/analog-discovery-studio-max/start>)
 - Analog Discovery Studio (<https://diligent.com/reference/test-and-measurement/analog-discovery-studio/start>)
 - Analog Discovery Pro 2000 Series (<https://diligent.com/reference/test-and-measurement/analog-discovery-pro-2230/start>)
 - Analog Discovery Pro 3000 Series (<https://diligent.com/reference/test-and-measurement/analog-discovery-pro-3x50/start>)
 - Analog Discovery Pro 5000 Series (<https://diligent.com/reference/test-and-measurement/analog-discovery-pro-5000-series/start>)
 - Digital Discovery (<https://diligent.com/reference/test-and-measurement/digital-discovery/start>)
 - Analog Discovery 2 (<https://diligent.com/reference/test-and-measurement/analog-discovery-2/start>)
 - Analog Discovery (Legacy) (<https://diligent.com/reference/test-and-measurement/analog-discovery/start>)
- A Computer with WaveForms Software Installed

Guide

1. Opening the Logic Analyzer

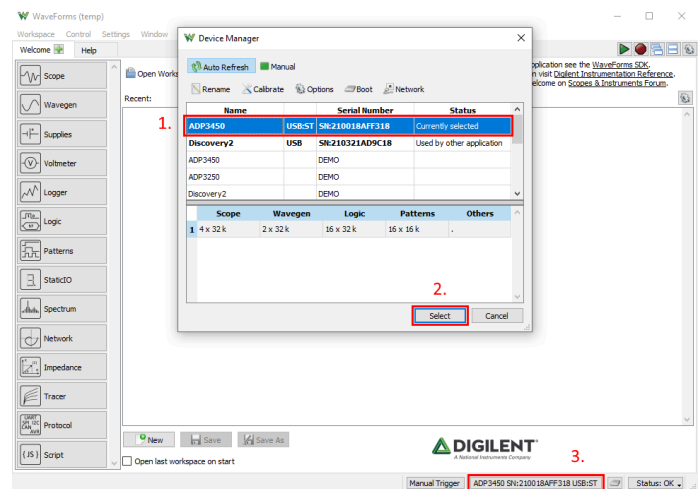
1.1

Plug in the Test & Measurement Device, then start WaveForms and make sure the device is connected.

If no device is connected to the host computer when WaveForms launches, the **Device Manager** will be launched. Make sure that the device is plugged in and turned on, at which point it will appear in the Device Manager's device list (1). Click on the device in the list to select it, then click the Select button (2) to close the Device Manager.

Note: “DEMO” devices are also listed, which allow the user to use WaveForms and create projects without a physical device.

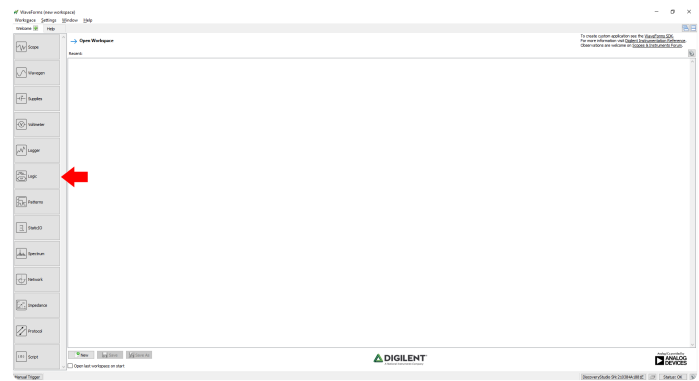
Note: The Device Manager can be reopened by clicking on the “Connected Device” button in the bottom right corner of the screen (3), or by selecting “Device Manager” from the “Settings” menu at the top of the screen.



(https://digilent.com/reference/_detail/reference/test-and-measurement/guides/waveforms-oscilloscope-device-manager.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

1.2

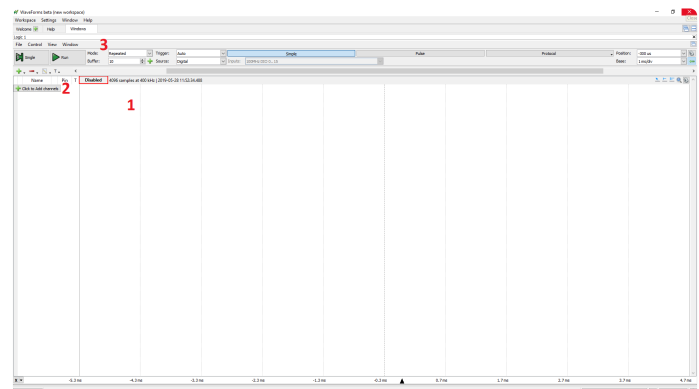
Once the Welcome page loads, in the instrument panel at the left side of the window, click on the **Logic** button to open the Logic Analyzer instrument.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/wave-instrument-panel/waveforms-instrument-panel-logic.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

1.3

Once the Logic Analyzer instrument opens, the window contains the data plot (1) showing captured data, the configuration panel (2) to left of the plot, and the control toolbar (3) at the top of the window.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/anal-discovery-studio-logic-analyzer/1.3.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

2. Using the Logic Analyzer

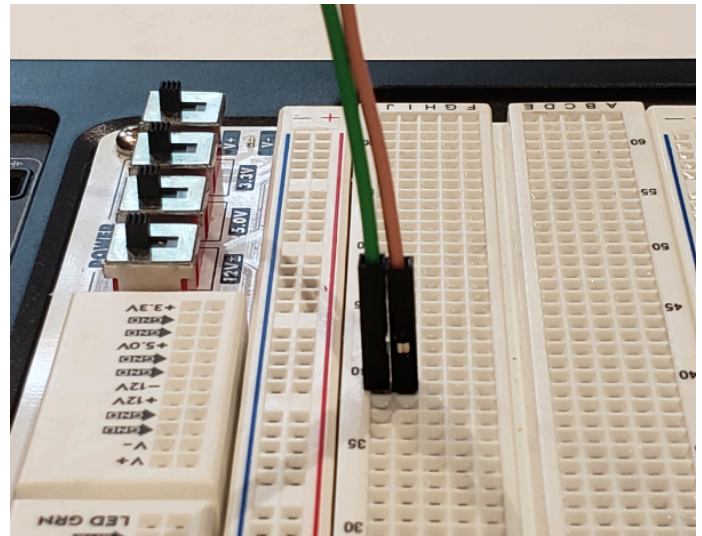
This section walks through setting up the Logic Analyzer to capture an incoming data signal

2.1 Hardware Setup

Choose two of your device's digital pins, one of which will act as an input, and one of which will act as output. Connect these two pins together to create a simple loopback, as seen in the

image to the right. Take note of which pins are used, as WaveForms must be told which pins it is to use.

Note: Digital pins that work as both input and output can be used in both modes at the same time. If desired, one such pin can be chosen for both input and output for this guide. In this situation, no additional hardware setup is required.



A  High Speed Adapter (<https://digilent.com/shop/digital-discovery-high-speed-adapter-and-logic-probes/>) is available for the Digital Discovery which can be used to take advantage of the highest sampling frequencies the device offers.

Click here if using the Digital Discovery High Speed Adapter

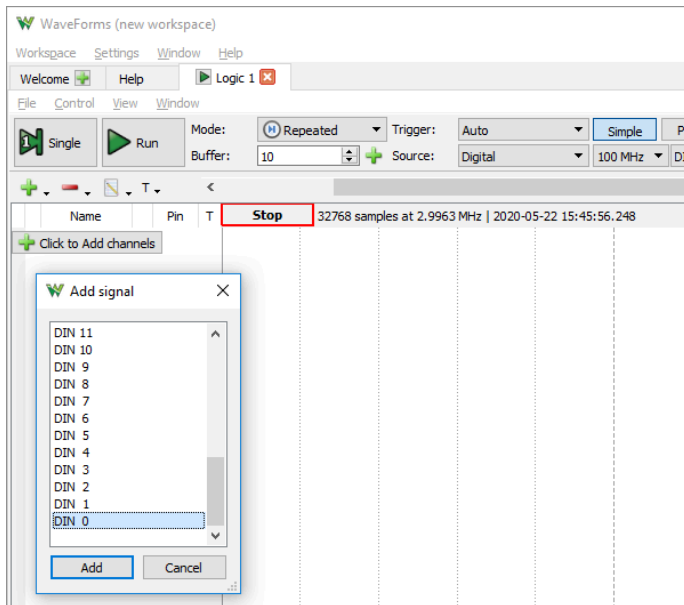
This adapter should be connected to the Digital Discovery's "HIGH SPEED INPUTS" pin group with the Ground symbols (downward facing arrows) pointing to the bottom of the device. While the adapter can be connected "upside down", doing so will result in the pin names shown in the [Digital Discovery Reference Manual](https://digilent.com/reference/test-and-measurement/digital-discovery/reference-manual#accessories) (<https://digilent.com/reference/test-and-measurement/digital-discovery/reference-manual#accessories>) not matching the actual pin names. Note that these pins cannot be used for digital output.



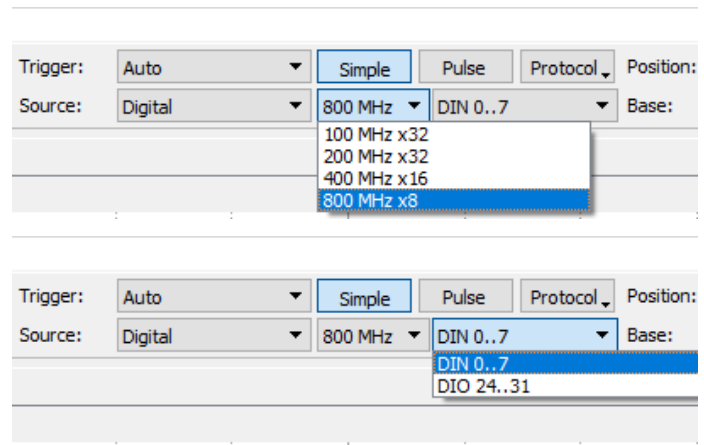
If desired, one of these input pins can be used in this guide, along with any of the Digital Discovery's output-capable pins. Connect one of the provided black/red differential probes to the High Speed Adapter. Make sure that the black cable is connected to the side with the ground arrow. The black cable should be connected to the ground of the device under test (in this case, the Digital Discovery's ground). The red cable (the positive side of the p/n differential pair) should be connected to the chosen digital output.

2.2 Software Setup

In the *Logic Analyzer* instrument that was previously opened, in the left panel, select "Click to Add channels" or the green plus button (+). The menu that pops up allows signals to be added as individual lines, a bus, a standard protocol, or a custom protocol. To add individual pins as their own lines, select "Signal" and use CTRL and/or SHIFT to select multiple pins from the list that pops up. For now, select only the pin you chose as input and click the **Add** button.

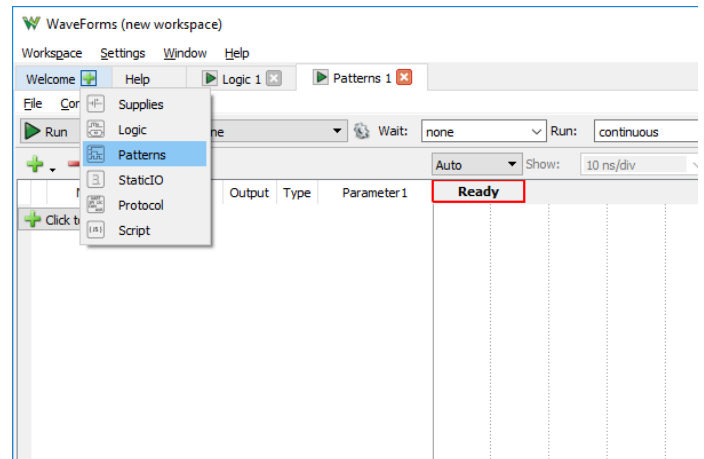


Next, check the sampling frequency which will be used for the digital input. Depending on the Test and Measurement device used, some frequencies may only be available with certain groups of pins. These frequency and pin requirements can be discovered through dropdowns in the Logic Analyzer instrument's toolbar, pictured to the right, as well as through the device's Reference Manual. If the device used does not support variable sampling frequencies, these dropdowns will be grayed out.

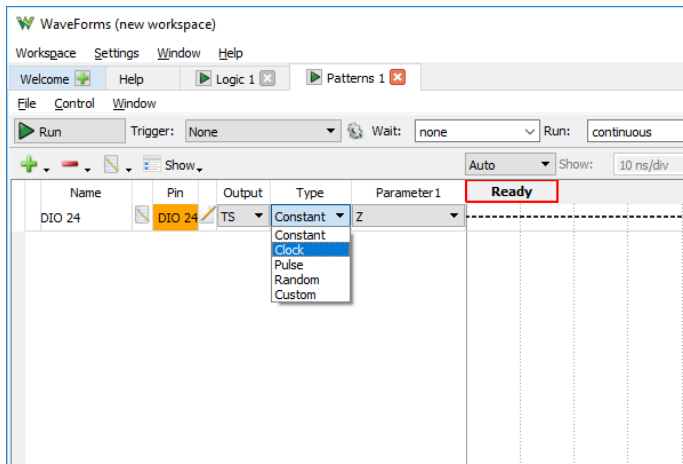


To use the *Logic Analyzer*, some signal/s must be applied to the digital input/s used. This guide uses the device's *Pattern Generator* instrument to apply a signal to the chosen digital input pin through the chosen digital output pin. For a detailed tutorial on using the *Pattern Generator* instrument, please see the [Using the Pattern Generator](https://digilent.com/reference/test-and-measurement/guides/waveforms-pattern-generator) (<https://digilent.com/reference/test-and-measurement/guides/waveforms-pattern-generator>) guide.

Return to WaveForms' Welcome page by clicking on its tab at the top left of the window and click the *Patterns* button. Alternatively, press the green plus button next to "Welcome" in the tabs list to add the instrument.



As before, in the left pane, click the "Click to Add channels" button or the green plus button (+). From "Signal" select the pin chosen as digital output and click the **Add** button. Under the **Type** column, choose *Clock* and leave the other settings in their default state.



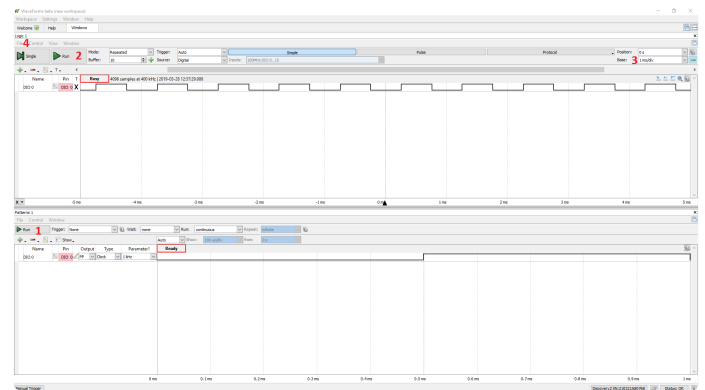
2.3 Capture Data

Click the *Pattern Generator's* Run button (▶) in the control bar (1 in the image to the right) to begin outputting the signal on digital input/output pin 0.

Return to the *Logic Analyzer* instrument, then click the Run button (2). The signal being generated by the *Patterns* instruments will appear in the plot.

Change the time scaling by selecting a different value in **Base** (3), located in the top right of the instrument (see [Section 3.4](#) for more on this).

To save the captured data, select “File” (4) from the top of the window and choose either “Save Acquisition” (allows for WaveForms to reopen the instrument with the data preloaded) or “Export” (allows export of a .csv file or a screenshot).

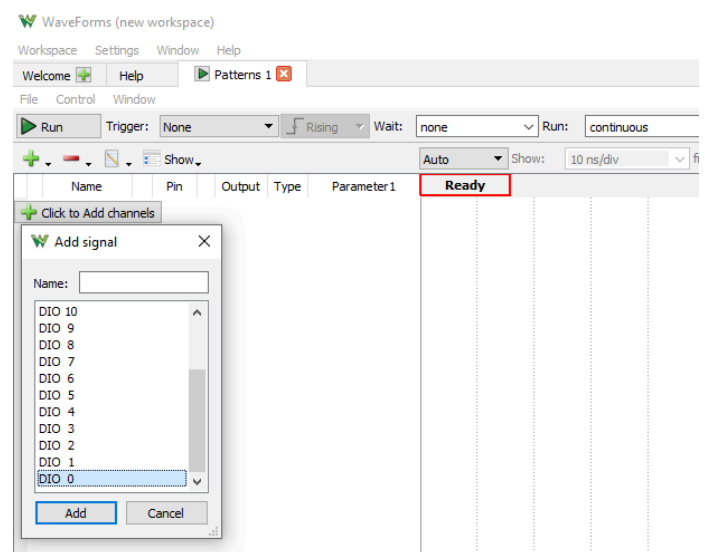


(https://diligent.com/reference/_detail/learn/instrumentation/tutorials/analysis-discovery-studio-logic-analyzer/2.2.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

Click here for an example of decoding a Digital Protocol

From the WaveForms home page, open the *Pattern Generator* instrument, then click on the **Click to Add channels** button.

From the list that opens, select **Signal**, then add **DIO 0**.

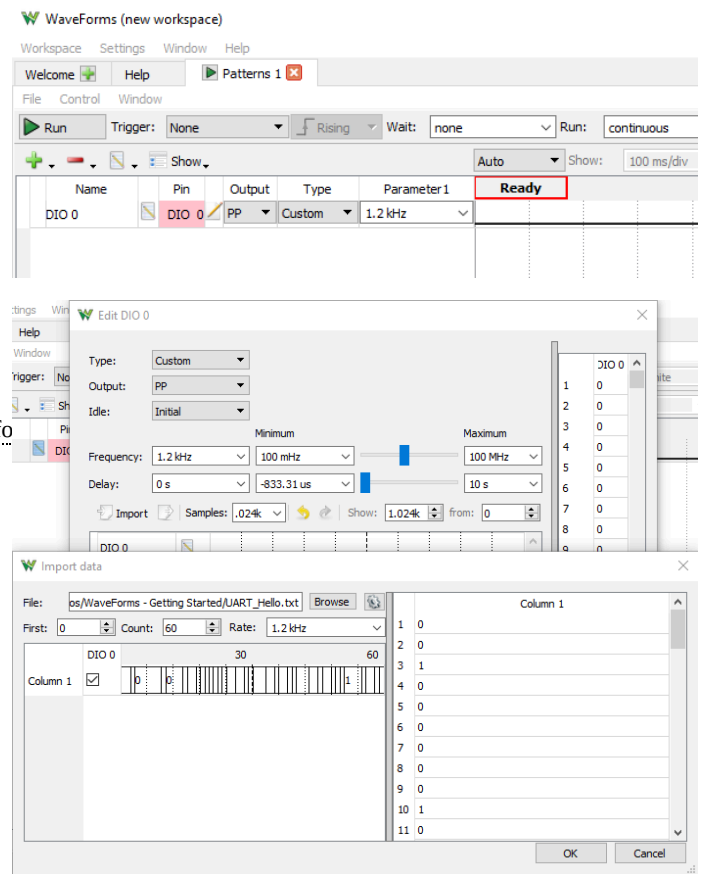


Set **Output** to **PP**, **Type** to **Custom** and **Parameter 1** to **1.2 kHz ()**, then open the signal editor (for more details see: [Using the Pattern Generator](#) (<https://diligent.com/reference/test-and-measurement/guides/waveforms-logic-analyzer>)).

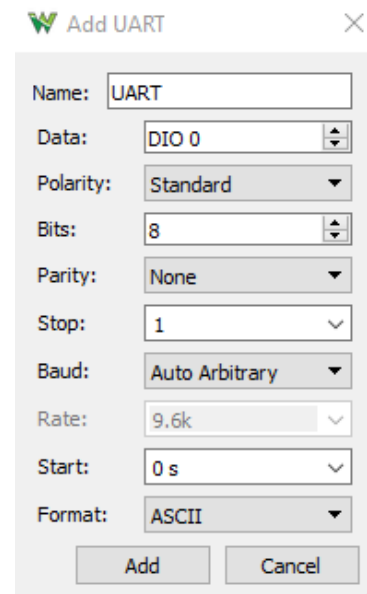
measurement/guides/waveforms-pattern-generator)).

In the signal editor, click on the **Import** button, select txt as file type then download, unzip and select the example file provided here: [uart_example.zip](https://digilent.com/reference/_media/learn/instrumentation/tutorials/waveforms-logic-analyzer/uart_example.zip)

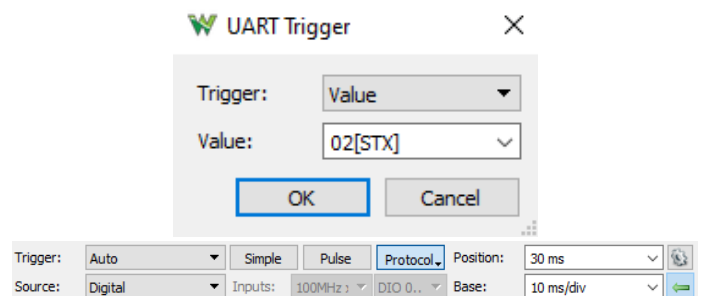
(https://digilent.com/reference/_media/learn/instrumentation/tutorials/waveforms-logic-analyzer/uart_example.zip). On the pop up window, leave everything as it is. Accept the changes and start the *Pattern Generator* with the **Run** button.



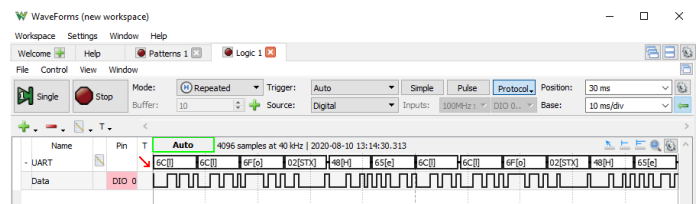
With the green plus button on the Welcome tab (+), open the *Logic Analyzer* instrument, then click on the **Click to Add channels** button. From the list that opens, select **UART**, then add **DIO 0** with the default settings.



At the trigger options, choose **Auto** and **Protocol**, select **UART** and set **Trigger** to **Value** and **Value** to **02[STX]** (start of text character), then accept the changes. Set the **Base** to **10 ms/div** and the **Position** to **30 ms** to make the message easily readable (You might need to play around with the Position value, to see the message).



Start the *Logic Analyzer* with the **Run** button to display the message. Once the UART transactions are generated by the *Pattern Generator*, and captured by the *Logic Analyzer*, you will be able to see the data captured. This includes both the input signal's waveform, as well as the data encoded in the transactions, as both hexadecimal numbers and the ASCII () representations.



Note: Variable Digital Supplies

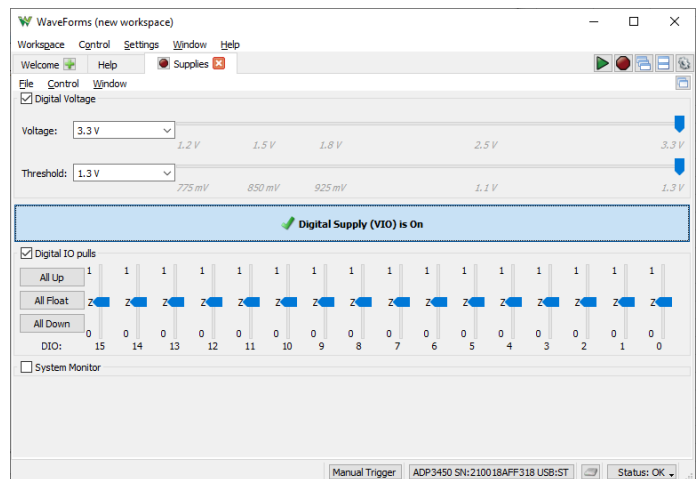
On some devices, the Analog Discovery Pro (ADP3450) and Digital Discovery, several additional configuration options for the digital I/O are included in the *Supplies* instrument.

As seen in the screenshot to the right, the supply voltage (and digital output voltage) can be changed with the first of two sliders. The threshold voltage of the digital inputs is also changed with this slider, as can be seen in the second of the two sliders. This feature allows the user to change the I/O standard used for the digital I/Os. It should be noted that the supply voltage and threshold are shared by all digital I/Os.

Additionally, the default pull of each signal can be assigned on a per-channel basis. When pull-ups or pull-downs are enabled an undriven pin will be pulled up to the supply voltage or down to ground, respectively.

Note: This screenshot was taken with an ADP3450 connected. There are some minor differences with the user interface for the *Digital Discovery*.

For more information, check out [Using the Power Supplies](https://diligent.com/reference/test-and-measurement/guides/waveforms-supplies) (<https://diligent.com/reference/test-and-measurement/guides/waveforms-supplies>).



(https://diligent.com/reference/_detail/reference/test-and-measurement/guides/digital-supplies.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

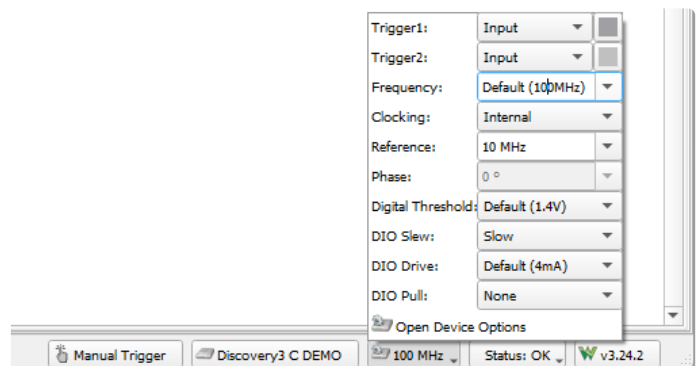
Note: System Frequency Adjustment

Many Diligent oscilloscopes feature the ability to select from a range of frequencies for the base system clock, which affects the sample rates of many instruments. Click the dropdown below to learn how to use this feature.

Modify the System Clock Rate

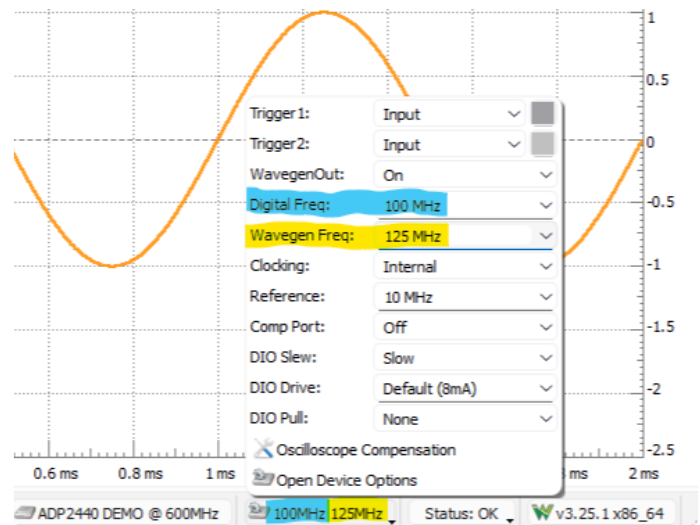
Clocking options are generally located in the application's footer, alongside other information about the status of connected devices. In this menu, the *Frequency* option sets the base system frequency. For details on other clocking options available for your device, refer to its reference manual, here on Reference.

Changing the system clock rate changes the set of sample rates that can be achieved by analog inputs, outputs, and digital I/O pins. Achievable rates for any particular setting are integer divisions of that system's sample rate. For example, for these devices, a 62.5 MS/s () sample rate is achievable at 125 MHz (), and a 25 MS/s () sample rate at 50 MHz () or 100 MHz ().



(https://diligent.com/reference/_detail/test-and-measurement/guides/waveforms-set-system-frequency.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

The Analog Discovery Pro 2440 and 2450 provide users with the ability to adjust the **Digital Freq** output frequency separate from the **Wavegen Freq** reference clock used by the AWG. In some cases, adjusting the clock rate used by the DIO can help users more cleanly generate output signals that have clearly defined duty cycles.



(https://digilent.com/reference/_detail/test-and-measurement/analog-discovery-pro-2000-series/wavegen125mhzclockref.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

Note: Using Multiple Instruments

The *Logic Analyzer* instrument can be used with other WaveForms instruments. Using multiple instruments at the same time is helpful in a wide variety of situations. One example would be to use the *Protocol Analyzer* (<https://digilent.com/reference/test-and-measurement/guides/waveforms-protocol-analyzer>) to configure a device under test, the *Pattern Generator* or *Protocol Analyzer* to stimulate the device, and capturing the result in the *Logic Analyzer*, all at the same time. More information on how instruments can be operated together can be found in the *Using Cross Triggering* (<https://digilent.com/reference/test-and-measurement/guides/waveforms-cross-triggers>) guide.

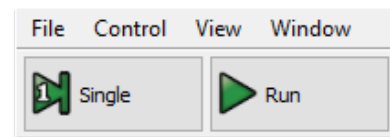
It should be noted that while multiple instruments can use any Digital Input pin, and those pins can be used as Digital Outputs, no more than one instrument can control a Digital Output pin at a time. WaveForms accounts for this by disabling instruments using a shared output, or by displaying a message stating that a shared output is currently used by a different instrument.

3. Logic Analyzer User Interface Overview

This section walks through the wide variety of controls and features available in the Logic Analyzer instrument

3.1 Control Buttons

The control bar near the top of the instrument window contains two buttons used to start/stop the capture of data. The Run button continues capturing data until it is stopped. The Single button will fill a buffer based around a pre-defined trigger (explained in following sections). The buffer size and position relative to the trigger are set by the **Position** and **Base** options (explained in Section 3.4).

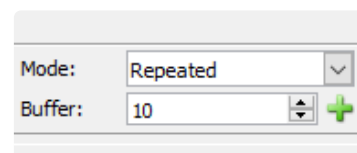


(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analog-discovery-studio-logic-analyzer/control_buttons.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

3.2 Buffer Configuration

To the right of the control buttons are various options for filling the buffer and selecting which buffer to look at. The **Mode** drop-down changes between the five acquisition modes: *Repeated*, *Screen*, *Shift*, *Record*, and *Sync*.

Record mode repeatedly fills buffers with data and displays them on the screen as they're filled until the Stop button is pressed.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analog-discovery-studio-logic-analyzer/buffer_control.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

Screen and *Shift* modes continuously add new data to the screen as it streams in instead of showing full buffers as in *Record*. *Screen* will fill the screen from left to right and then add new data on the left, overwriting the previous data set. *Shift* will show new data as it comes in from the right, in effect scrolling the data from right to left.

Record allows for large amounts of data to be captured at a lower sample rate. This can be used for recording data from a sensor that needs to be monitored for several minutes or hours.

Sync offers the ability to use an external clock to control when the *Logic Analyzer* records samples. Useful with communication protocols such as SPI, I2C, and UART.

The **Buffer** option allows for selection of buffers, as each acquisition fills a new one (except in *Repeat* mode, where buffers are sequentially filled). The green plus button (+) creates tabs for each buffer for easy switching.

3.3 Trigger Configuration

The **Trigger** drop-down allows for selection of the trigger type: *None*, *Auto*, and *Normal*.

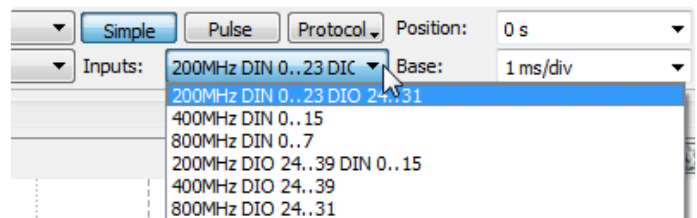
None starts the acquisition immediately after Run is clicked.

Auto starts the acquisition either on the trigger event, or approximately 2 seconds after Run is pressed if the trigger event doesn't occur.

Normal starts the acquisition only on the trigger event.

The **Source** drop-down allows the user to trigger the oscilloscope based on events in other instruments.

The **Inputs** drop-down (pictured on the right) is available for Diligent's **Digital Discovery** to specify which DIO pins to use at a specified frequency.



(https://diligent.com/reference/_detail/learn/instrumentation/tutorials/anal-discovery-studio-pattern-generator/dd_sample_rate.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

The three buttons labeled “Simple”, “Pulse”, and “Protocol” allow for customization of trigger conditions.

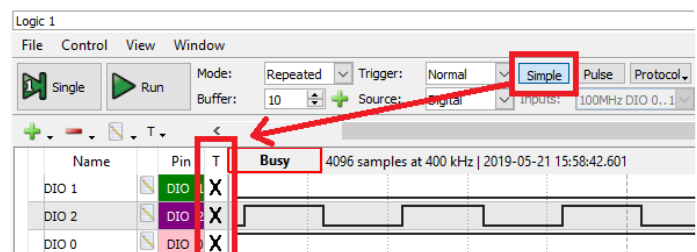
Simple is used in conjunction with the Trigger column (labeled T in the *Logic Analyzer* instrument, see image at right). Each channel can be set to trigger on various events: Ignore (X), Low (0), High (1), Rise (↗), Fall (↘), and Edge (↕).

Pulse opens a menu to configure a trigger on a pulse that is either shorter than a specified time (**Glitch**), longer than a specified time (**Timeout**), the exact length of specified time (**Length**), or after a specified number of edges (**Counter**).

Protocol is used in conjunction with a Bus configuration or one of the standard protocols (SPI, I2C, UART, CAN, I2S, and 1Wire) available when adding channels to the *Logic Analyzer* instrument. A protocol-specific menu will pop up to configure the trigger.

3.4 Time Configuration

By default, the Time group in the Configuration panel contains the **Position** and **Base** fields.



(https://diligent.com/reference/_detail/learn/instrumentation/tutorials/anal-discovery-studio-logic-analyzer/trig_config.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

The **Position** field centers the plot on the selected time, measured from the trigger. This can be used to view data captured before and after the trigger event.

The **Base** field configures the scale used for the horizontal axis of the plot.

Using these two settings can be thought of as “panning” and “zooming” the plot.

The gear button (⚙️) opens a menu (pictured on the right) that can be used to modify the Position and Base field units, and the acquisition behavior.

Position as a division toggles the position units between seconds and divisions.

Range Mode drop-down has the options *Full*, *Division*, and *PlusMinus*.

Full changes the units in Base to the full length of the buffer in seconds.

Division changes the units in Base to the number of seconds per division (note there are always 10 divisions in the *Logic Analyzer* instrument).

PlusMinus changes the units in Base to the number of seconds before and after the Position point.

The **Clock** drop-down changes between the *Internal* and *External* clocks. This option is only available for certain pieces of hardware, and therefore may be grayed out.

Noise toggles the option to acquire noise samples in half of the buffer.

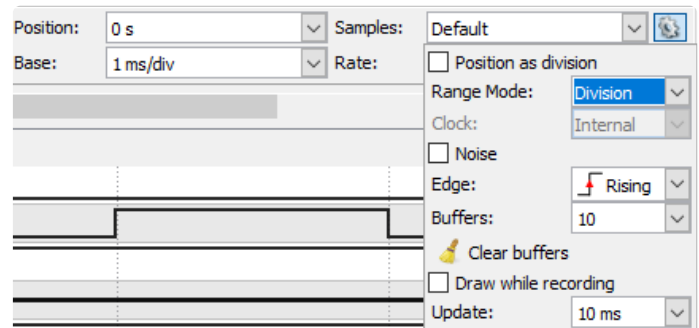
The **Edge** drop-down allows for selection of the rising, falling, or edge for the external or internal trigger signals.

The **Buffers** drop-down selects the number of buffers to store in the PC. This changes the maximum value in **Buffer** in Section 3.1.

The “Clear buffers” button (🗑️) clears all the PC buffers.

Draw while recording toggles drawing while in *Record* and *Sync* modes, somewhat resembling *Scan* and *Screen* behavior while collecting data over a long period

The **Update** drop-down sets the update rate of the *Logic Analyzer* instrument display.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analog-discovery-studio-logic-analyzer/time_config.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

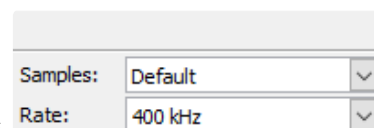
3.5 Additional Configuration Options

Pressing the green arrow (➡️)

(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analog-discovery-studio-logic-analyzer/arrow.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer) below the gear icon will display options for setting the number of samples to collect and sample rate. Changing these options overrides the Base selections

Samples selects the number of data points to collect.

Rate selects the sampling rate. Together with the values selected in the Samples drop-down, this will determine the Base value.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analog-discovery-studio-logic-analyzer/addtnl_config.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)

3.6 Adding Protocol Channels

Protocol channels can be used to aid interpretation of digital communication protocol transactions. For example, SPI, UART, I2C, and other protocol channels can be used to represent data words in a variety of different formats (including binary, decimal, hexadecimal, ASCII (), and many others). Data words are aligned with the appropriate flags (start, select, acknowledge, etc), based on user-defined settings.

In order to add a protocol channel to the plot, follow the steps in Section 2.1 of this guide and select the appropriate protocol from the list. This will cause a menu with settings specific to that protocol to pop up, allowing for selection of DIO channels for each signal line. Protocol-specific options such as endianness, polarity, bit order, etc can also be defined.

After being added to the plot, protocol channel settings can be modified by clicking the Properties icon (🔧) for that channel (1 in the image at right).

The Protocol and Bus modes will display the data seen on each individual line, as well as the overall value of that combination of channels. These values are marked by 2's in the image at right.

3.7 View Options

The **View** menu is located at the top of the *Logic Analyzer* instrument window, right above Run. Selecting any of the views in that menu will open a tabbed window on the right side of the instrument window. The available views are *Data*, *Events*, *Logging*, *Measurements*, *Cursors*, and *Notes* (see image at right).

Data displays a table of all acquired samples. If using the Bus or a protocol, only the overall values (as discussed in 3.5) are displayed.

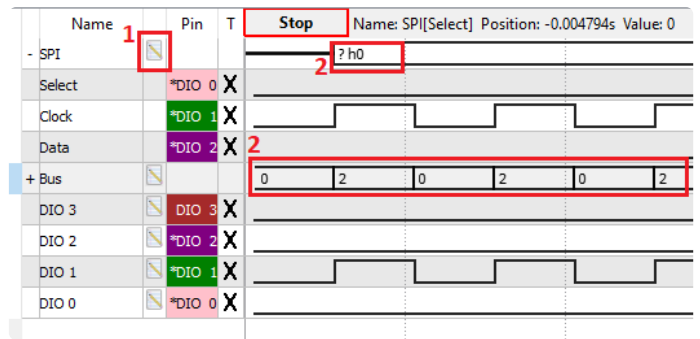
Events only displays data when it changes. This provides a more compact view than Data, but only displays one channel at a time. Channels can be selected in the drop-down presented at the top of the Events view, as well as filtered by various methods.

Measurements shows the list of selected measurements. These available measurements are cycles, frequency, period, positive/negative duty cycles, and positive/negative pulse widths.

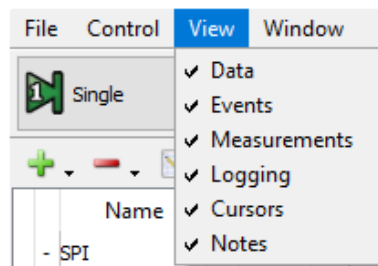
Logging uses scripts to automate the capture of multiple data sequences, and save the captured data to file/s.

Cursors are used to display time reference markers with respect to the capture-triggering event. The drop-down menus contain adjustment controls for the position, reference cursor, and delta x value.

Notes allows for adding of notes and descriptions into projects.



(https://digilent.com/reference/_detail/learn/instrumentation/tutorials/analysis-discovery-studio-logic-analyzer/protocol_channels.png?id=test-and-measurement%3Aguides%3Awaveforms-logic-analyzer)



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Next Steps

For more guides on how to use the Digilent Test & Measurement Device, return to the device's Resource Center, linked from the [Test and Measurement](https://digilent.com/reference/test-and-measurement/start) (<https://digilent.com/reference/test-and-measurement/start>) page of this wiki.